

DACH Region Transport Performance and Cost Audit Report

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05-2026

1. Business Context

The main goal of this quarterly report is evaluation of delivery timeliness from Poland to Germany and Austria and comparison of freight costs on specific routes.

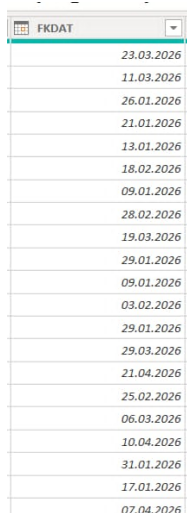
2. SAP S/4HANA Simulation

The data used in this report come from SAP S/4HANA, module SD and TM.

Name	Description
VBELN	Billing document
TOR_ID	Freight order
MANDT	Mandant number
FKDAT	Document date
LIFNR_CARRIER	Vendor ID
CARRIER_NAME	Carrier name
FREIGHT_COST	Freight cost in EUR
NETWR_BILLING	Value of goods in EUR
ON_TIME_FLAG	Delivered on time
IN_FULL_FLAG	Delivered in full
OTIF_COMPLIANT	Delivered on time in full

3. ETL in PowerQuery

In order to show the proper data, the data was transformed. FKDAT COLUMN has been changed to type date. NETWR_BILLING and FREIGHT_COST have been transformed into numeric type of column.



The screenshot shows a table with a single column labeled 'FKDAT'. The column contains a list of dates in the format 'DD.MM.YYYY'. The dates are: 23.03.2026, 11.03.2026, 26.01.2026, 21.01.2026, 13.01.2026, 18.02.2026, 09.01.2026, 28.02.2026, 19.03.2026, 29.01.2026, 09.01.2026, 03.02.2026, 29.01.2026, 29.03.2026, 21.04.2026, 25.02.2026, 06.03.2026, 10.04.2026, 31.01.2026, 17.01.2026, and 07.04.2026.

FKDAT
23.03.2026
11.03.2026
26.01.2026
21.01.2026
13.01.2026
18.02.2026
09.01.2026
28.02.2026
19.03.2026
29.01.2026
09.01.2026
03.02.2026
29.01.2026
29.03.2026
21.04.2026
25.02.2026
06.03.2026
10.04.2026
31.01.2026
17.01.2026
07.04.2026

1.2 NETWR_BILLING	1.2 FREIGHT_COST
13492,39	1087,51
3506,15	665,89
9569,71	760,22
4676,41	788,94
15942,94	1082,26
7370,8	1064,7
9634,65	1094,69
9587,13	835,09
15619,1	1314,56
12022,98	1334,16
10617,75	1107,34
11068,74	1230,68
4589,84	695,45
12392,07	842,38
8515,19	1392,39
14221,18	1126,91
13194,6	1269,47
8223,3	1112,08
12860,24	1015,95
14754,39	1032,56
15477,77	1105,86

DELIVERY_STATE column has been added as a conditional column.

Dodawanie kolumny warunkowej

Dodaj kolumnę warunkową obliczaną na podstawie innych kolumn lub wartości.

Nazwa nowej kolumny

DELIVERY_STATE

	Nazwa kolumny	Operator	Wartość		Wartość wyjściowa
Jeśli	OTIF_COMPLIANT	równa się	0	To	non-compliant

Dodaj klauzulę

W przeciwnym razie

compliant

OK

Anuluj

Column FREIGHT_COST_RATIO has been added, it is a percentage division result freight cost and value of goods.

Kolumna niestandardowa

Dodaj kolumnę obliczaną na podstawie innych kolumn.

Nazwa nowej kolumny

FREIGHT_COST_RATIO

Formuła kolumny niestandardowej ⓘ

= [FREIGHT_COST]/[NETWR_BILLING]

Dostępne kolumny

- NETWR_BILLING
- FREIGHT_COST
- WAERK
- ON_TIME_FLAG
- IN_FULL_FLAG
- OTIF_COMPLIANT
- DELIVERY_STATE

<< Wstaw

Więcej informacji o formułach programu Power Query.

✓ Nie wykryto błędów składniowych.

OK

Anuluj

ABC 123 FREIGHT_COST_RATIO WŁAŚC

- 1.2 Liczba dziesiętna
- \$ Stałoprzecinkowa liczba dziesiętna
- 1²3 Liczba całkowita
- % Wartość procentowa
- Data/godzina
- Data
- Godzina
- Data/godzina/strefa czasowa
- Czas trwania
- A^BC Tekst
- ✓ Prawda/fałsz
- Binarny
- Używając ustawień regionalnych...

4. Data Visualization in PowerBI

The following charts were used in PowerBI:

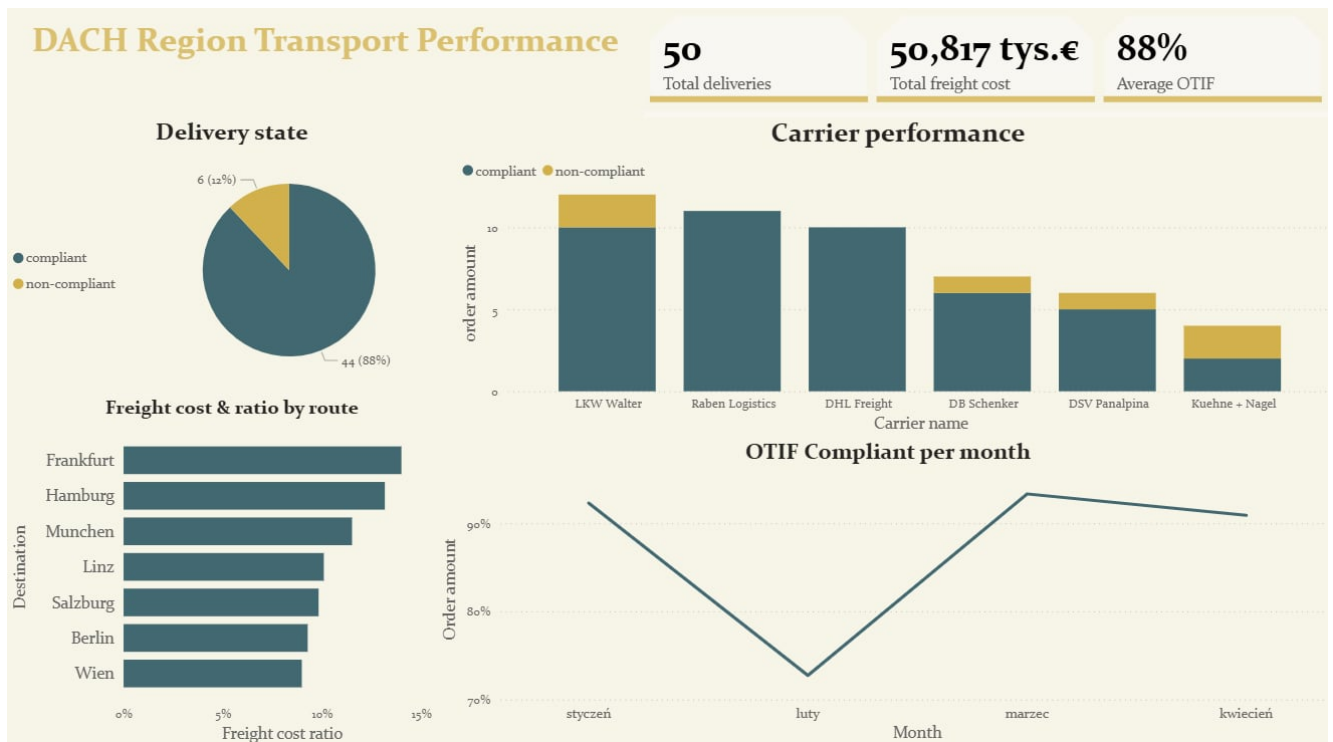
Cards: Total deliveries, Total freight cost and Average OTIF

Donut chart – Delivery state

Bar chart – Carrier performance

Horizontal bar chart – Freight cost & ratio by route

Trend chart – OTIF Compliant per month



5. Conclusions

The overall delivery timeliness (OTIF) stands at 88%. The total freight cost is 50 820 EUR. While LKW Walter handles the highest volume of orders, Raben Logistics achieved a flawless 100% on-time delivery rate. Conversely, Kuehne + Nagel generated the highest percentage of delayed (non-compliant) shipments. The route to Frankfurt is the least cost-efficient, with transport costs consuming the highest percentage of the goods' net value (highest Freight cost ratio). Deliveries to Vienna remain the most profitable in terms of freight optimization. Supply chain reliability fluctuates seasonally. The highest OTIF compliance was recorded in March, whereas the lowest performance occurred in February.